CIRPIN: searching beyond sequence order

Systematic discovery of circular permutations across the protein universe using CIRPIN

Aiden R. Kolodziej, S. Mazdak Abulnaga, Sergey Ovchinnikov

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The question: which structural relatives do searches miss?

- Proteins can preserve their 3D arrangement while moving the N and C termini.
- Sequential alignment then sees a broken correspondence; even graph embeddings can encode sequence order too strongly.
- Can a fast structure embedding ignore this specific rearrangement while retaining useful structural discrimination?

Contribution
Synthetic circular permutations train an invariant search model; disagreement with an order-sensitive model nominates candidate rearrangements.

Introduction; Fig. 1.

Circular permutation is a structural relationship

Original: $N\!-\!A-B-C-D-C$

Connect old termini, cut after B

Permuted: $N\!-\!C-D-A-B-C$

The blocks retain their 3D arrangement and cyclic order. Only the chain's starting point changes.

Crucial distinction
This definition does not establish common ancestry. Convergence and divergence require additional evolutionary evidence.

Defining Circular Permutation; schematic is explanatory, not an experimental protein.

A concrete failure: the same coordinates, a different score

QZ (d1qzya1) and K8 (d2dk8a1) share a circularly permuted winged helix–turn–helix fold.

- Progres: 0.389, below its 0.8 similarity threshold.
- CIRPIN: 0.979.
- Rotating QZ's residue order alone can rescue the Progres match.

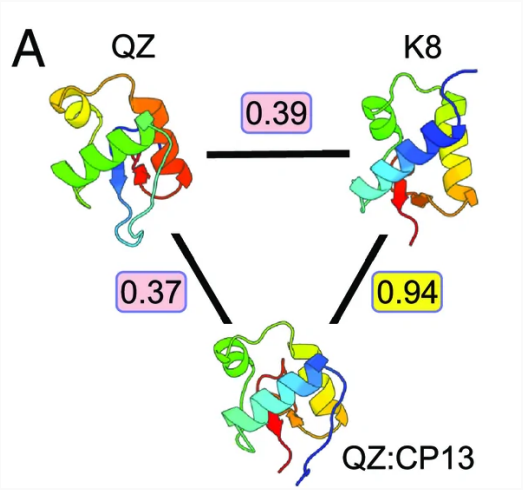


Fig. 2A (cropped); Table 2. Scores quoted from Table 2.

Inputs and outputs: a structure becomes a searchable vector

1. Input: a single protein-domain structure, represented by Cα coordinates.
2. Build a spatial graph: connect residues within 10 Å.
3. Equivariant graph neural network updates residue representations; mean pooling yields a domain embedding.
4. Output: rank database structures by cosine similarity of embeddings.

$$\text{score}(a, b) = \frac{z_a^\top z_b}{\|z_a\| \, \|z_b\|}$$

Database embeddings are precomputed. The initial search does not explicitly align residue pairs.

Methods: Protein Structure Representation; CIRPIN Architecture and Training; Fig. 1C.

Keep positional information; teach it which changes to ignore

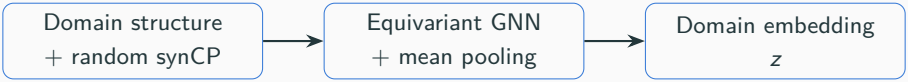
Each residue carries normalized local density, an N/C-terminal indicator, and a 64-dimensional sinusoidal positional encoding. For coordinates $\mathcal{P} = (p_1, \ldots, p_n)$, augmentation at cut k is

$$\mathcal{P}_k = (p_{k+1}, \ldots, p_n, p_1, \ldots, p_k).$$

- The 3D coordinates stay fixed; residue indices and terminal features change.
- Random cuts expose the model to many orderings across training epochs.
- These synthetic examples need not be physically realizable proteins: no folding or linker design is performed.

Methods: Synthetic Circular Permutation. Removing positional encoding entirely harms retrieval (prior Progres work).

Training changes the equivalence relation



Same-family embeddings pulled together;
different-family embeddings pushed apart.

Supervised contrastive learning brings proteins in the same SCOPe family together and separates other families, while random synCPs remove dependence on the chain start. Trained for 500 epochs; model selected using the Progres validation protocol. This is learned robustness, not an architectural proof of exact invariance.

Fig. 1; Methods: CIRPIN Architecture and Training. Diagram redrawn for explanation.

Does invariance collapse unrelated structures?

- QZ/K8: CIRPIN similarity remains high across all synthetic cut combinations; Progres scores vary widely.
- Unrelated comparison: CIRPIN remains low across cuts, whereas Progres again varies.
- Ordinary SCOPe retrieval remains comparable, with small losses:

Model	Fold	Superfamily	Family
CIRPIN	0.166	0.692	0.865
Progres	0.177	0.706	0.877

These are reported retrieval scores, not CP precision estimates.

Fig. 2C; Table 1. Test: 400 domains; no training domain at $\geq$30% sequence identity to test cases.

A small curated CP benchmark gives a clear gain

Method	All 11 known pairs	Excluding 2 same-family pairs
Progres	4/11	2/9
CIRPIN	9/11	7/9
TM-align	3/11	—
TM-align with -cp	7/11	—

- Similarity thresholds: embedding cosine > 0.8; TM-score > 0.5.
- Same-family pairs are especially favorable because training explicitly pulls them together.
- Two CIRPIN misses also fail TM-align -cp: additional structural changes complicate a pure CP match.

Fig. 2D; Table 2. Eleven literature-derived pairs verified visually; a small, heterogeneous benchmark.

Two scores nominate; explicit alignment adjudicates

Define $\Delta CP = s_{\mathrm{CIRPIN}} - s_{\mathrm{Progres}}$.

1. High CIRPIN, low Progres: similar geometry that sequence-order features obscure.
2. Align with and without CP: TM_{cp} and TM_{seq}.
3. Require global similarity and improvement when wrapping the chain:

$$TM_{cp} > 0.5, \qquad \Delta TM = TM_{cp} - TM_{seq} > 0.$$

Use the smaller of query- and target-length-normalized TM-scores, so a tiny shared fragment is insufficient.

Interpretation
High ΔCP alone does not identify a CP: rewiring, insertions and false positives also produce disagreement.

Fig. 3; Methods: Structural Alignment with TM-Align. ΔTM is defined here explicitly to avoid sign ambiguity.

SCOPe: most embedding disagreements fail structural validation

Stage	Count
SCOPe40 domains	∼15,200
Pairs: CIRPIN > 0.9, Progres < 0.6	18,326
Pairs failing $TM_{cp} > 0.5$	16,249 (∼89%)
Pairs classified as likely CP	1,973
Unique domains in those CP pairs	1,204

Failures often share secondary-structure content: about 80% share a SCOPe class, versus a roughly 20% random baseline. Architecture-level resemblance is insufficient.

Prevalence and Discovery of Circular Permutations in SCOPe; Fig. 3.

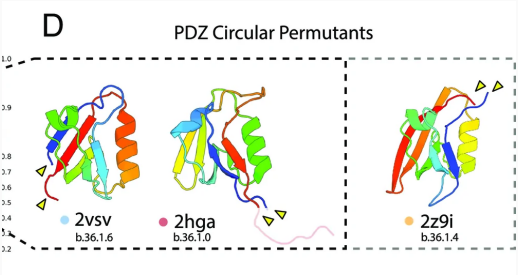
Prevalence is sensitive to redundancy and ascertainment

- Clustering the SCOPe CP domains gives 193 structural clusters, compared with 4,357 across all SCOPe40.
- Varying embedding cutoffs by ± 0.05 gives 193–304 clusters: an estimated 4.4–7.0% cluster-level prevalence.
- Prior CPDB has repeated concanavalin-like structures ($\sim$20% of pairs), low-TM candidates, and engineered permutants.

How to read the estimate
It is a threshold- and database-dependent fraction of structural clusters, not the fraction of all proteins in nature that underwent a CP event.

SCOPe prevalence and CPDB bias sections; supplement sensitivity/quality-control analyses described in main text.

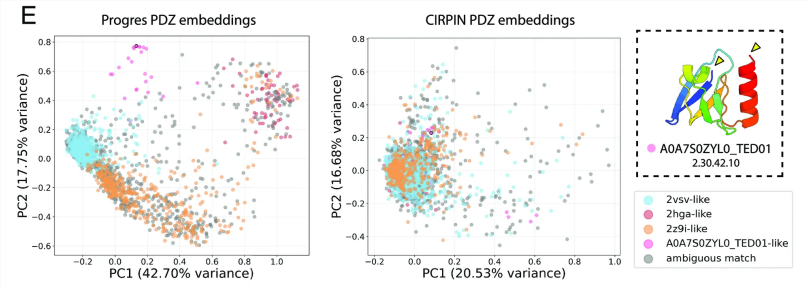
PDZ: a known fold conceals multiple chain starts



- SCOPe search highlights the PDZ domain-like fold (b.36).
- Three topological forms include the previously known 2z9i permutant and the newly recognized 2hga form.
- Removing 2hga's C-terminal tail does not rescue Progres similarity, arguing against a tail-only explanation.

Fig. 4D (cropped); Novel Circular Permutations of PDZ Domains. N-to-C coloring retained; triangles mark termini.

The fourth PDZ form emerges in predicted structures



4,015 TED-annotated PDZ domains → 2,563 at 75–100 residues. A separate Progres cluster contains a fourth CP topology; CIRPIN brings the forms together. PCA visualizes the embeddings; the topology claim rests on structural comparison. Labels are ambiguous when reference TM-scores differ by < 0.025.

Fig. 4E (cropped; axes and legend retained). Pink fourth form: A0A7S0ZYL0_TED01.

Scaling up: a fast filter still leaves enormous alignment work

AFDB-ClustR pipeline	Count
Representative predicted proteins	$\sim$2.7 million
TED-parsed domains	$\sim$3.5 million
CIRPIN $>$ 0.8 and ΔCP cutoff 0.5	584,583,567 pairs
After $TM_{cp} > 0.5$, $\Delta TM > 0$	18,301,678 pairs
Unique domains	589,113
Structural clusters	44,875

Lowering ΔCP to 0.4 yields 8.9 billion candidate pairs. The cutoff trades sensitivity to subtle CPs against feasible validation cost.

Identifying Novel CPs in AFDB-ClustR; Methods: AFDB-ClustR Threshold Determination; Table S1 described in main text.

18.3 million pairs are enriched candidates, not a gold standard

Manual audit: 100 pairs

72 confident CPs

12 unclear

16 unrelated by CP

- Reported estimated precision: 72%, including TM-align filtering.
- Audit spans over 100 unique CATH annotations, but remains a small sample.
- Database contains 845 annotated CATH topologies; many pairs lack annotation.

Pair counts grow with family redundancy; neither 18.3 million pairs nor 589,113 domains counts independent evolutionary events.

AFDB-ClustR results; Fig. 5A. Interpretation of independence is journal-club commentary.

Case studies support structural discovery, with distinct evidence levels

- Endopeptidase examples corroborate previously described CP relationships.
- New ribonuclease H-like domain pairs illustrate discovery beyond the curated SCOPe search.
- Cross-comparisons to T2-like/BECR RNases have low core TM-scores (0.38, 0.27); visual motifs motivate hypotheses, not established ancestry.

Evidence boundary
Predicted structures and shared topology generate evolutionary hypotheses. Sequence analysis, phylogeny and functional experiments are needed to test mechanism and biological equivalence.

Fig. 5B–E; AFDB-ClustR results. Qualification distinguishes structural evidence from evolutionary inference.

The model also retrieves changes that are not circular permutations

- Rewiring: secondary-structure elements occupy similar places but have different internal connectivity; a single circular cut cannot explain the relationship.
- Insertions/extensions: deleting the inserted region rescues a representative Progres score; CIRPIN changes little.
- Such pairs can have high ΔCP but $\Delta TM = 0$.

Why this matters
Model disagreement detects a broader class of order-sensitive retrieval failures. Explicit topology inspection gives the biological interpretation.

Figs. 7–8; CIRPIN Retrieves Protein Pairs Obscured by Rewiring and Insertions.

What do residue embeddings suggest the model learned?

- 1uur: CIRPIN node similarities reflect three comparable helices.
- QZ: N- and C-terminal beta-sheet residues show similarity despite their sequence separation.
- 3cbn: two interlocking, superimposable substructures (TM-score 0.81) show an internal similarity signal absent from Progres.

Authors' interpretation
Augmentation reduces overreliance on absolute position and encourages local structural-context representations.

These are selected examples; they do not prove that every embedding feature has a motif-level interpretation.

Fig. 9A–C; Tertiary Motifs Emerge from CIRPIN Node Embeddings.

Limitations that change how we would use CIRPIN

1. **Detection bias:** large score disagreements favor severe CPs; same-family relationships may score highly in both models and be filtered out.
2. **Generalization:** training is based on SCOPe; distant folds and inaccurate domain boundaries remain problematic.
3. **Validation:** the known-CP benchmark is small; the manual audit finds a meaningful false-positive fraction.
4. **Evolution:** small 60–80-residue folds with 4–6 secondary elements are especially vulnerable to convergence-based interpretations.

Discussion; benchmark and AFDB-ClustR results. No claim of genome-wide CP recall is supported.

Discussion: a useful new search axis, not the final evolutionary answer

1. For our protein of interest, would an order-invariant hit reveal a missing structural relative? Inspect the alignment and domain boundaries first.
2. What benchmark would distinguish CP robustness from generic tolerance to structural changes? Use independently curated, held-out topologies with negative controls.
3. For the PDZ forms, which experiment best tests a functional consequence: folding stability, peptide specificity, or insertion tolerance?

Take-home

Synthetic rearrangements make a fast embedding search less sensitive to chain start. Combining invariant and order-sensitive scores enables large-scale candidate discovery, followed by structural and evolutionary scrutiny.

Journal-club interpretation and discussion prompts; not additional experimental results.

Sources and reproducibility

- Journal article: Kolodziej, Abulnaga & Ovchinnikov. *PNAS* 123(36), e2605178123 (2026). Published online September 4, 2026. `https://doi.org/10.1073/pnas.2605178123`
- Full text and figure source (CC BY 4.0): `https://pmc.ncbi.nlm.nih.gov/articles/PMC13552848/`
- Code/model resources: `https://github.com/aidenkoloj/CIRPIN`
- Data: `https://huggingface.co/datasets/aidenkzj/CIRPIN`

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